

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

DEEP LEARNING

(Common to IT, AI&DS, CSE (DS), CSE, CSE (CS) and CS&D)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

- (a) Find the eigen values for the matrix $A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$. 2M
- (b) What is single value decomposition of a matrix? 2M
- (c) What is bias and variance in machine learning? 2M
- (d) Differentiate supervised Learning with unsupervised Learning. 2M
- (e) What is tangent classifier? 2M
- (f) Define L1 regularization. 2M
- (g) What is convolution operation in CNN? 2M
- (h) The output layer of convolutional network is usually relatively inexpensive to learning layer. Why? 2M
- (i) Illustrate the block diagram of LSTM. 2M
- (j) List the advantage of recursive nets over recurrent nets. 2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 (a) Explain the following information theory concepts briefly. 5M
(i) Entropy, (ii) Information gain.
- (b) What is gradient based optimization? Why is it needed? Explain. 5M

OR

- 3 Explain about principal Component analysis and its use in machine Learning. 10M

- 4 Explain various Gradient Based optimization techniques. 10M

OR

- 5 (a) Explain about XOR in machine Learning. 5M
- (b) What is Back propagation? Explain the working of Back propagation. 5M

- 6 Explain various bagging and boosting ensemble techniques. 10M

OR

- 7 What is adversarial training and adversarial attacks algorithms? Explain. 10M

- 8 Explain various components of Convolutional Neural Networks. How does a convolutional layer and pooling layer work? 10M

OR

- 9 (a) What are the variants of Convolutional Neural Networks? 3M
- (b) Explain Tiled convolution in detail. Write mathematical expression for the Tiled convolution. 7M

- 10 Illustrate with an example of Encoder- Decoder or sequence-to-sequence RNN Architecture. 10M

OR

- 11 (a) Write a short note on Gated RNNs. 5M
- (b) Explain about Bidirectional RNNs. 5M

B.Tech III Year I Semester (R20) Regular & Supplementary Examinations November/December 2024

DEEP LEARNING

(Common to CSE (AI) and CSE (AI&ML))

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

- (a) Illustrate how eigenvalues and eigenvectors are used in principal component analysis. 2M
- (b) Determine the expectation of a random variable given its probability distribution function. 2M
- (c) Explain the difference between supervised and unsupervised learning with examples. 2M
- (d) Describe the role of hidden units in determining the network's ability to learn complex patterns in the context of neural networks. 2M
- (e) Describe the concept of multi-task learning and its benefit in machine learning models. 2M
- (f) Describe the role of learning rate in gradient-based optimization algorithms for training deep neural networks. 2M
- (g) Outline the differences between fully connected and convolutional layers in terms of parameter efficiency. 2M
- (h) Identify the components of a convolutional network that contribute to handling spatial hierarchies in images. 2M
- (i) Describe how backpropagation through time works in the context of training recurrent neural networks. 2M
- (j) Identify the key differences between standard RNNs and bidirectional RNNs. 2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2 Using a dataset represented by a matrix A of dimensions 3×3: 10M

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

- (i) Perform an Eigen decomposition of the matrix. Explain the significance of eigenvalues and eigenvectors in understanding the properties of A.
- (ii) Suppose a random variable X follows a binomial distribution with parameters n=10 (number of trials) and p=0.5 (probability of success). Calculate the probability that X equals 5?

OR

- 3 (a) Explain the concept of gradient descent, including its mathematical foundation. 5M
- (b) Describe how gradients are used to minimize a loss function, detailing the steps involved in a single iteration of gradient descent. 5M
- 4 (a) Explain the role of validation sets in preventing under fitting and over fitting. 5M
- (b) Discuss how validation sets are used to tune the hyper parameters effectively. 5M

OR

- 5 (a) Describe the architecture of a deep feedforward network, including the roles of the input layer, hidden layers, and output layer. 5M
- (b) Explain how the depth and width of the network can impact its learning capacity and generalization abilities. 5M

Contd. in Page 2

- 6 Explain the concept of dataset augmentation and its importance in training deep learning models? Discuss various techniques of data augmentation for both image and text data? 10M

OR

- 7 (a) Describe the concept of parameter initialization and its significance in neural network training. 5M
(b) Explain the effects of different initialization strategies on the convergence and performance of the model. 5M
- 8 (a) Discuss the impact of different parameters in the convolution operation, such as kernel size, padding, and stride. 5M
(b) Explain how these parameters influence the output feature map and the network's ability to detect spatial hierarchies in data. 5M

OR

- 9 (a) Explain how different data types influence the design and functionality of convolutional networks. 5M
(b) What is the role of efficient convolution algorithms in processing these data types? 5M
- 10 Describe the architecture of a bidirectional RNN. Explain how data flows through this type of network and compare it to the flow in traditional unidirectional RNNs. 10M

OR

- 11 Describe the architecture of a basic auto encoder. Explain the process of encoding and decoding in an auto encoder and how these operations enable the network to learn efficient data representations? 10M

B.Tech III Year II Semester (R20) Supplementary Examinations November/December 2024

DEEP LEARNING

(Artificial Intelligence and Machine Learning)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|---|----|
| (a) What are the basic components of linear algebra? | 2M |
| (b) Define Eigen decomposition. | 2M |
| (c) What is under fitting in machine learning? | 2M |
| (d) What is the significance of back-propagation in deep feed forward networks? | 2M |
| (e) What is dataset augmentation? | 2M |
| (f) Define semi-supervised learning. | 2M |
| (g) What is the convolution operation in convolutional networks? | 2M |
| (h) What is the efficient convolution algorithms used in practice? | 2M |
| (i) What are recurrent neural networks (RNNs)? | 2M |
| (j) Define bidirectional RNNs. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- | | | |
|-----------|--|----|
| 2 | (a) How do matrices and tensors differ from each other? | 5M |
| | (b) Explain how singular value decomposition is used in principal components analysis. | 5M |
| OR | | |
| 3 | (a) Compare and contrast conditional probability and marginal probability. | 5M |
| | (b) What are the different types of norms used in linear algebra? | 5M |
| 4 | (a) Discuss the process of learning XOR in deep feed forward networks. | 5M |
| | (b) Explain how bias and variance impact the performance of machine learning models. | 5M |
| OR | | |
| 5 | (a) Compare and contrast maximum likelihood and Bayesian statistics. | 5M |
| | (b) Discuss the challenges that motivate the adoption of deep learning techniques. | 5M |
| 6 | (a) Compare and contrast bagging and ensemble methods. | 5M |
| | (b) Discuss the challenges faced in neural network optimization. | 5M |
| OR | | |
| 7 | (a) Discuss the different regularization techniques used in deep learning. | 5M |
| | (b) Explain the concept of parameter tying and sharing in regularization. | 5M |
| 8 | (a) Discuss the role of convolutional networks in deep learning. | 5M |
| | (b) Describe random or unsupervised features in convolutional networks. | 5M |
| OR | | |
| 9 | (a) What are the applications of convolutional networks in real-world scenarios? | 5M |
| | (b) Explain the importance of basis for convolutional networks in feature extraction. | 5M |
| 10 | (a) Explain the concept of auto encoders in sequence modeling. | 5M |
| | (b) Describe the optimization techniques used for long-term dependencies in sequence modeling. | 5M |
| OR | | |
| 11 | (a) What are the applications of sequence modeling in natural language processing? | 5M |
| | (b) Explain the concept of recursive neural networks and their applications. | 5M |

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